

3. RL

Environment Design

State Representation (20 dimensions)

The state combines two sources

- **15 PCA components** the number recommended in the Dimensionality Reduction work .
- **5 binary domain flags** the EDA's top discriminators by F-statistic:
has_prior_denial (F=5742), has_customer_value (F=1884),
is_fast_claim (F=2044), coverage_limit_high (F=1542),
is_zero_touchpoints (F=796). These are appended raw because their signal is so specific that PCA would dilute it.

Action Space (4 actions)

- 0 :Fast-track Approve
- 1 :Request Documentation
- 2 :Manual Review
- 3 : Deny

Reward Function

- Correct approve (label=1, action=0): +10
- Correct deny (label=0, action=3): +8
- Wrong fast-track (label=0, action=0): $-20 - 5 \times (1-p)$
- Wrong deny (label=1, action=3): -3
- Delay (action=1 or 2): -1 + uncertainty bonus

The $-5 \times (1-p)$ term where p is the supervised model's probability is the key **reward shaping** mechanism that solves the flat-reward problem.

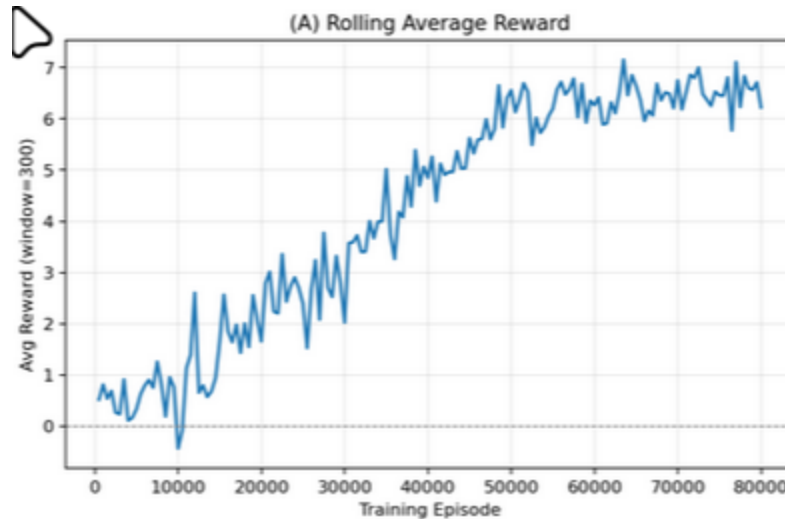
Q-Learning Agent

Each of the 20 continuous state dimensions is quantised into 5 equal-frequency bins using training-set quantiles. States are stored as a sparse hash map only ~9,025 unique states were actually visited from a theoretical maximum of 5^{20} .

Training schedule:

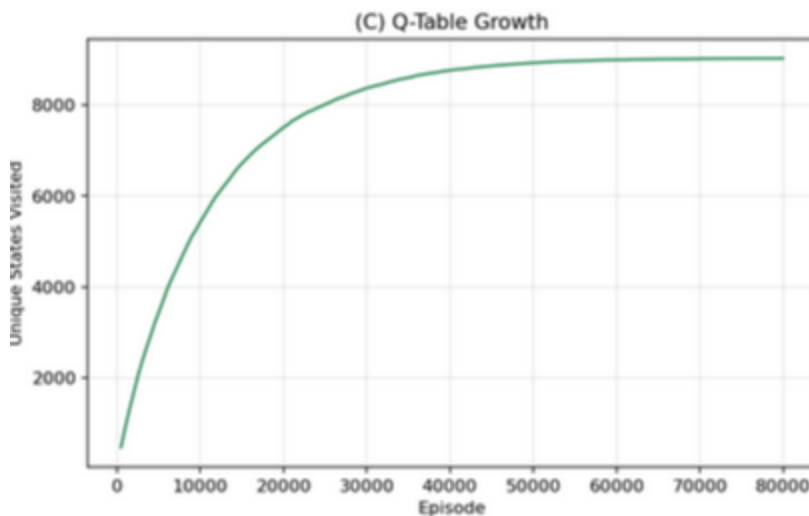
- ϵ greedy with linear decay from 1.0 $\rightarrow$ 0.05 over 50,000 steps
- Cosine-annealing learning rate: $\alpha = 0.35 \rightarrow 0.05$ (prevents late-training oscillations)
- 80,000 total episodes

Learning curve behaviour



The agent starts with negative average reward (random exploration), then improves steadily from episode ~15,000 as ϵ drops and exploitation increases. By episode 50,000 ($\epsilon=0.05$) the rolling reward stabilises around +6–7 on training data.

The Q-table size saturates at ~9,025 states, confirming the discretisation converged.



Evaluation & Comparison

Policy	Avg Reward	Accuracy	AUC	F1 (Class 0)	Risky Claims Fast-Tracked
Q-Learning	+2.39 ✓	0.75	0.52	0.12	557
Always Approve	2.24	0.76	0.50	0.00	705
Supervised RF	1.10	0.76	0.51	0.07	24
Random	0.38	0.63	0.50	0.23	191
Always Review	0.07	0.76	0.50	0.00	0

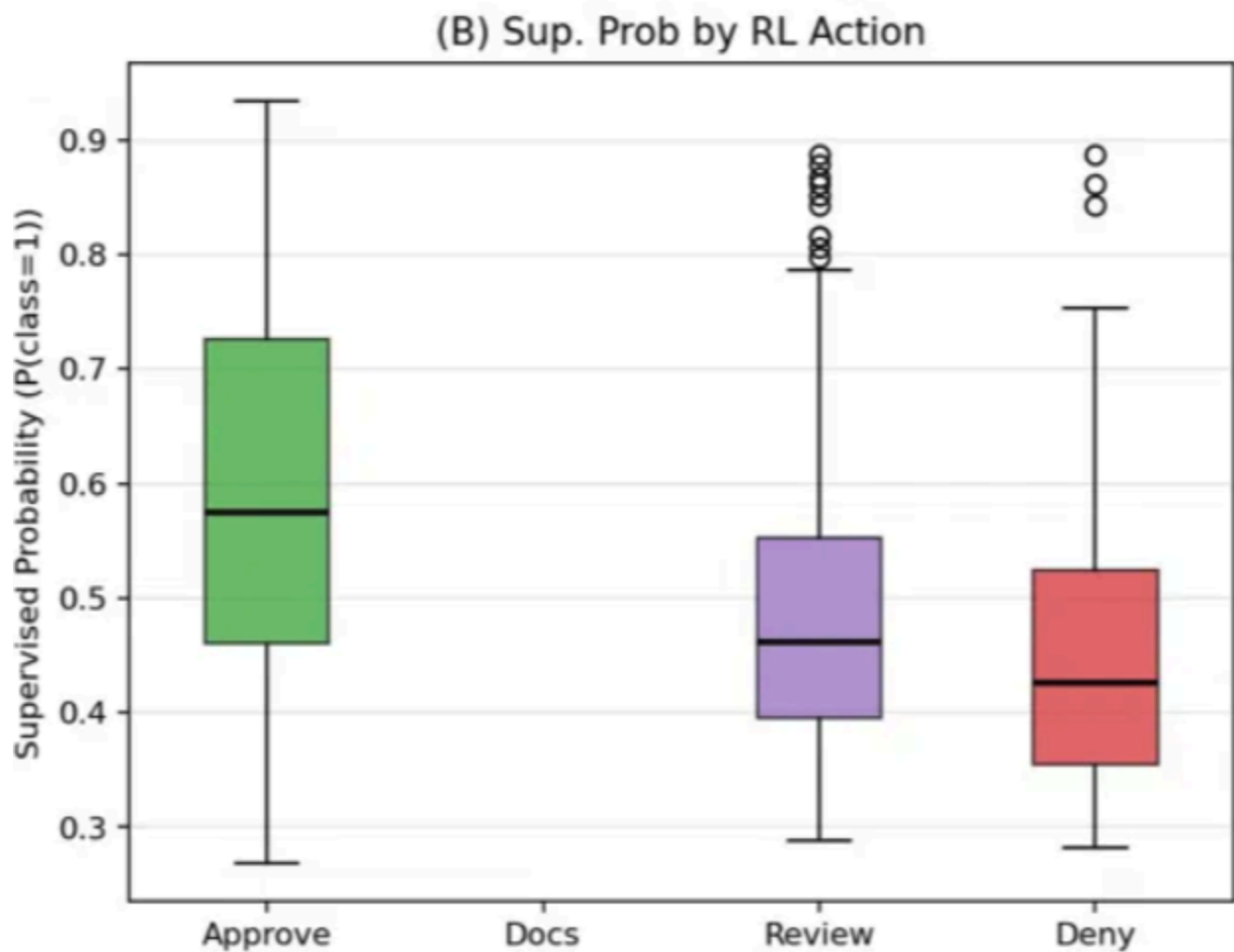
Q-Learning achieves the highest average reward (+2.39), outperforming even Always Approve (+2.24) despite similar accuracy.



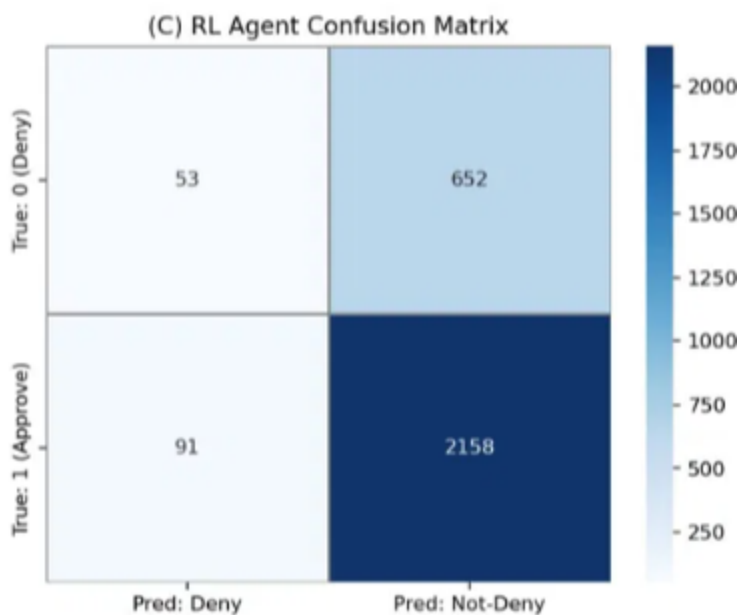
Decision Pattern Analysis

- The **supervised probability boxplot** confirms the agent does differentiate: claims it sends to review or deny have meaningfully lower P(approve) than those

it fast-tracks



- The **confusion matrix** shows the agent is aggressive on approvals, conservative on denials



The Critical Trade-off

The **risky fast-track count** exposes the fundamental RL limitation here: Q-Learning fast-tracks **557 risky claims** vs. only **24** for the supervised RF. The RF is dramatically safer because it was trained to discriminate classes. The RL agent instead optimised for reward, and since 76% of claims genuinely deserve approval, a mostly-approve policy earns high cumulative reward while still missing substantial fraud.

RL vs. Supervised Learning

The supervised RF optimises **classification**. it minimises misclassification symmetrically, hence excellent fraud detection. The RL agent optimises **cumulative reward**. it learned that the reward-maximising policy under this specific function is close to "approve most, deny the clearly suspicious." This makes RL more flexible (the policy adapts if the reward function changes) but also more dangerous if fraud prevention is the primary business objective. The practical recommendation is a **hybrid**: use RF probability as part of the state (as done here via reward shaping), and deploy RL as the decision layer to optimise for operational efficiency on top of the supervised signal.